

WireGuard VPN Deployment Guide

Purpose

WireGuard provides secure, encrypted remote access for analysts, administrators, and students connecting to the Virtual SOC environment. Unlike floating IP or RDP-based access, WireGuard allows: - Direct access to internal SOC hosts - Encrypted tunnels into pfSense - Role-based peer management - No exposure of internal networks to the public Internet

WireGuard is deployed directly on pfSense and provides a stable, secure operator endpoint.

1. Install WireGuard on pfSense

pfSense GUI:

```
System → Package Manager → Available Packages  
Search: wireguard  
Install
```

Once installed:

```
VPN → WireGuard → Settings  
Check: Enable WireGuard  
Save
```

2. Create WireGuard Tunnel

Navigate:

```
VPN → WireGuard → Tunnels → + Add Tunnel
```

Configure: - **Listen Port:** 51820 - **Generate Keys:** Click "Generate Keypair" - **Address:** 10.5.5.1/24

This assigns pfSense as the WireGuard gateway for peers.

Save tunnel.

3. Assign WireGuard Interface

Navigate:

Interfaces → Interface Assignments

A new interface (e.g., `wg0` or `OPT1`) appears in the list. Add interface, then configure: - **Static IPv4:** 10.5.5.1/24 - **Enable interface**

Save & apply changes.

4. Add WireGuard Peers

Each analyst or administrator must be assigned a unique peer block.

Example addressing:

```
10.5.5.3/32
10.5.5.4/32
10.5.5.5/32
10.5.5.6/32
```

Step 1 — Generate Local Peer Keys

On local workstation:

```
wg genkey | tee wg-private.key | wg pubkey > wg-public.key
```

Step 2 — Add Peer to pfSense

Navigate:

VPN → WireGuard → Peers → + Add Peer

Enter: - **Public Key:** peer's public key - **Allowed IPs:** <peer IP>/32 - **Endpoint:** Leave blank (dynamic peers)

Save.

5. Firewall Rules

WireGuard requires two rule sets.

WAN Rule — Allow WG Handshake

```
Firewall → Rules → WAN  
Add → UDP → Port 51820 → Destination: WAN address → Allow
```

WireGuard Interface Rule — Allow Peer Traffic

```
Firewall → Rules → wg0  
Pass: Protocol: Any → Source: WireGuard net → Destination: Any
```

This authorizes full tunnel or split tunnel depending on peer config.

6. (Optional) NAT Rules for Full Tunneling

If peers must route all traffic through pfSense:

```
Firewall → NAT → Outbound  
Add: Source = WireGuard net → Interface = WAN
```

Most VSOC setups use **split tunnel** (internal subnets only).

7. Local Peer Configuration

Edit peer config:

```
sudo nvim /etc/wireguard/wg0.conf
```

Example template:

```
[Interface]  
Address = 10.5.5.<peer-ip>/32  
PrivateKey = <local-private-key>
```

```
[Peer]
PublicKey = <pfsense-wireguard-public-key>
Endpoint = <pfsense-wan-ip>:51820
AllowedIPs = 10.0.0.0/8
PersistentKeepalive = 25
```

Bring up the tunnel:

```
sudo wg-quick up wg0
```

Verify:

```
wg
```

Shows handshake timestamps and transfer counters.

8. Connectivity Validation

From Peer

```
ping 10.5.5.1      # pfSense WG gateway
ping 105.45.5.1    # pfSense LAN
ping 105.45.5.72   # Internal hosts
```

Access Wazuh Dashboard

```
https://<wazuh-ip>:443
```

Only reachable through the tunnel.

9. Operational Usage

WireGuard enables: - Secure SOC operator access - Remote lab access for students - Secure management plane for SIEM and pfSense - Access to all internal systems without exposing them publicly

Peer keys and addresses are easily revoked or rotated for access control.

10. Summary of Responsibilities

WireGuard provides: - Encrypted VPN access into VSOC environment - Dedicated operator subnet (10.5.5.0/24) - Authentication via cryptographic keys - Secure, low-overhead tunneling

This access method replaces the temporary RDP-based GUI access and secures the operational endpoint for all future SOC users.